
Voice-Preserving Multilingual Video Translator Using NLP and AI Voice Cloning (G47)

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Abstract

Multilingual speech translation remains a challenging problem due to data imbalance, linguistic variability, and the inherent difficulties in preserving speaker identity and audiovisual synchronization. In this work, we implement a modular, end-to-end pipeline for high-fidelity German-to-English video dubbing, extending the principles of the DeepMind multilingual dubbing framework. We propose a robust architecture that replaces traditional components with state-of-the-art models: Whisper-Medium for noise-resilient transcription, NLLB-200 for high-accuracy machine translation, and Coqui XTTS-v2 for zero-shot voice cloning. To address the critical challenge of cross-lingual duration mismatch, we introduce a custom Elastic Duration Aligner that dynamically synchronizes translated audio with the original video timeline. Finally, visual coherence is achieved by fine-tuning a Wav2Lip GAN on high-resolution (256×256) facial crops from the MuAViC dataset. We evaluate our system using a suite of objective metrics, including Structural Similarity (SSIM), Fréchet Inception Distance (FID), and Landmark Distance (LMD), demonstrating a scalable and accessible approach to automated video dubbing.

1 Motivation and Objectives

1.1 Motivation

Linguistic barriers persist as a significant impediment to the equitable dissemination of information, particularly in online education and digital media. Current statistics indicate that while over half of Internet content is published in English, a vast majority of global users are non-native speakers. This disparity restricts access to high-quality educational resources for non-English populations.

Traditional automated dubbing solutions often suffer from the "uncanny valley" effect, where the translated audio desynchronizes from the speaker's lip movements, degrading viewer engagement. Furthermore, conventional pipelines that rely on cascaded ASR and MT systems often propagate errors, leading to semantic drift. By integrating Whisper-Medium, a robust, multitask model trained on 680,000 hours of diverse audio, our system reduces the reliance on massive, language-specific aligned datasets, thereby improving scalability for under-represented languages. Our project focuses on German-to-English dubbing as a proof-of-concept for a scalable, inclusive architecture that benefits educators, content creators, and global learners by democratizing access to video content.

1.2 Objectives

The primary objective of this work is to engineer and evaluate an end-to-end multilingual dubbing pipeline that seamlessly translates spoken content from German to English. We aim to overcome the limitations of traditional dubbing by ensuring both semantic accuracy and audiovisual coherence. Specifically, our goals are defined as follows:

- Quantitative Objectives: We aim to outperform or match the state-of-the-art baseline established in the "Large-Scale Multilingual Audio Visual Dubbing" framework. Specifically, we target:
 - An FID (Fréchet Inception Distance) score lower than 78.5, indicating high visual realism.
 - An SSIM (Structural Similarity) score ≥ 0.970 , ensuring high-fidelity frame reconstruction.
 - A PSNR (Peak Signal-to-Noise Ratio) greater than 31.38 dB.
 - Precise temporal alignment where translated audio segments strictly adhere to the original speaker's timestamps.
- Qualitative Objectives:
 - Speaker Identity Preservation: Successfully clone the timbre and prosody of the German speaker into the generated English audio using zero-shot synthesis.
 - Lip Synchronization: Generate visually natural mouth movements that are perceptually synchronized with the synthetic English phonemes, minimizing the jarring "dubbing effect."

2 Related Work and Background

2.1 Literature Review

Automated Video Dubbing Architectures: The field of automated video dubbing has evolved through two primary paradigms: cascaded systems and end-to-end architectures. Large-scale multilingual dubbing was pioneered by Yang et al. [2020], who proposed a modular cascaded pipeline integrating ASR, Machine Translation (MT), and Text-to-Speech (TTS). While their Residual U-Net architecture for lip synchronization set a benchmark for visual fidelity, their analysis revealed significant language bias [2020]. In contrast, Hu et al. introduced *Neural Dubber* [2021], an end-to-end model utilizing Image-based Speaker Embeddings (ISE) to clone timbre directly from facial features. More recently, Sung-Bin et al. proposed *VoiceCraft-Dub* [2025], leveraging Neural Codec Language Models (NCLMs) for high-expressivity synthesis, though such models often require prohibitive computational resources.

Duration Alignment: A critical challenge in cross-lingual dubbing is the duration mismatch caused by varying information densities across languages. Wu et al. addressed this with *VideoDubber* [2022], which integrates duration-aware positional embeddings into the NMT stage. Similarly, Chadha et al. [2025] proposed a Length-Sensitive Speech Translation (LSST) model using Length-Aware Beam Search (LABS). Recent work by Cui et al. [2025] frames duration alignment as a preference optimization problem (SSPO), highlighting that precise temporal alignment remains an open bottleneck.

2.2 Background: Evaluation Metrics

To rigorously assess the performance of our pipeline, we employ a suite of five objective metrics implemented in our evaluation framework:

- Fréchet Inception Distance (FID): A standard metric for generative models that measures the distance between the distribution of generated face images and ground truth frames in the feature space of an Inception-v3 network. Lower FID scores indicate higher realism Heusel et al. [2017].
- Structural Similarity Index (SSIM): A perceptual metric that quantifies the degradation in visual quality by comparing luminance, contrast, and structure between the generated and

ground truth frames. Values range from -1 to 1, with 1 indicating perfect similarity Wang et al. [2004].

- **Peak Signal-to-Noise Ratio (PSNR):** Measures the ratio between the maximum possible power of a signal (image pixel value) and the power of corrupting noise (reconstruction error). Higher dB values indicate better reconstruction fidelity.
- **Landmark Distance (LMD):** To evaluate lip-sync precision specifically, we calculate the Mean Euclidean Distance between the facial landmarks of the generated mouth region and the ground truth, as extracted by a face alignment network Chen et al. [2018].
- **Variance of the Laplacian (VLap):** A measure of image sharpness calculated by taking the variance of the convolution of the image with a Laplacian operator. This metric helps detect blurring artifacts common in GAN-generated faces Pech-Pacheco et al. [2000].

3 Methodology

Our proposed system is a modular, end-to-end pipeline composed of two primary branches: an Audio Processing branch responsible for translation and voice cloning, and a Visual Processing branch responsible for high-fidelity lip synchronization.

3.1 Dataset and Preprocessing

We utilize the German subset of the MuAViC (Multilingual Audio-Visual Corpus) dataset. To ensure high-fidelity visual synthesis, we implement a robust preprocessing pipeline:

- **Face Detection and Alignment:** We employ the S3FD (Single Shot Scale-invariant Face Detector) to locate faces in the source video frames. Unlike standard approaches that crop to 96×96 , we compute an affine transformation matrix to align and crop faces to a high resolution of 256×256 , preserving fine-grained facial details.
- **Audio Processing:** Source audio is extracted at 16kHz. For the visual model input, we generate Mel-spectrograms with a window size of 800, hop size of 200, and 80 Mel filter banks.
- **Information Leak Prevention:** During training, we apply a geometric mask to the lower half of the target face frame. This forces the generative model to infer lip movements solely from the audio input and the upper-face context, preventing it from "cheating" by copying pixel information from the ground truth.

3.2 The Audio Dubbing Pipeline

The audio branch operates sequentially to transform German speech into English while preserving the original speaker’s identity.

1. **Source Separation and Transcription:** We utilize Demucs to separate the source audio into vocal and accompaniment (background) tracks. The isolated vocals are transcribed using Whisper-Medium, which provides both text and precise timestamp boundaries $[t_{start}, t_{end}]$ for each utterance.
2. **Translation and Synthesis:** The German transcripts are translated to English using NLLB-200. We then employ Coqui XTTS v2 for zero-shot text-to-speech synthesis, conditioning the generation on the original German vocal track to clone the speaker’s timbre.
3. **Elastic Duration Alignment:** A core contribution of our work is the custom *Duration Aligner* designed to resolve the length mismatch between German and English speech. Let T_{src} be the duration of the original German segment and T_{gen} be the duration of the synthesized English audio. We calculate the ratio $r = T_{gen}/T_{src}$ and apply the following logic:

- **Compression:** If $r > 1.1$, the English audio is too long. We apply a non-pitch-shifting time compression with a speedup factor $S = \min(r, 1.25)$, capping acceleration at 25% to prevent artifacts.
- **Smart Positioning:** If $r < 1.0$, the audio is shorter than the visual window. We apply *Center-Alignment* (padding silence equally) for standard speech, or *Right-Alignment* for the first segment to strictly avoid overlapping with introductory silence or applause.

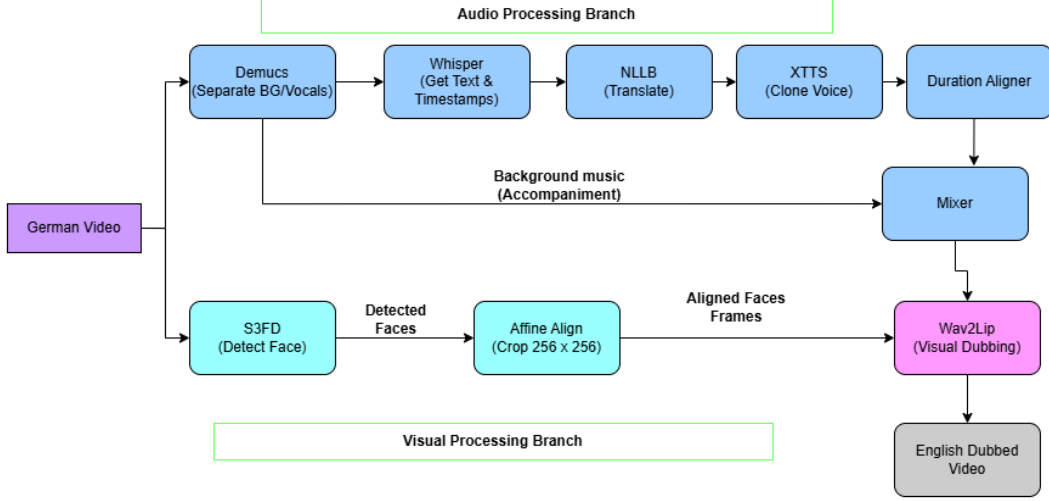


Figure 1: The proposed end-to-end German-to-English dubbing pipeline. The upper branch handles audio translation and elastic alignment, while the lower branch prepares visual features for the Wav2Lip generator.

3.3 Visual Synthesis Model

Our visual generator is based on the Wav2Lip architecture, modified to accept high-resolution inputs (256×256). The model is a U-Net style generator composed of three sub-networks:

1. Identity Encoder: A stack of 2D-Convolutional layers that encodes a random reference frame and the masked target frame. Input shape: $(B, 6, 256, 256)$.
2. Audio Encoder: A 2D-Convolutional stack processing the Mel-spectrogram input. Input shape: $(B, 1, 80, 16)$.
3. Face Decoder: A Deconvolutional network with skip connections that receives the concatenated audio and visual embeddings to reconstruct the mouth region.

Table 1 summarizes the component specifications.

Table 1: Model Architecture and Parameter Counts

Component	Input Shape	Params (M)
Visual Encoder	$(B, 6, 256, 256)$	14.2
Audio Encoder	$(B, 1, 80, 16)$	4.8
Face Decoder	$(B, 1024, 1, 1)^\dagger$	14.6
Discriminator	$(B, 3, 256, 256)$	13.9

[†] Bottleneck shape before upsampling.

3.4 Training and Optimization

We fine-tune the model using a weighted combination of three loss functions to ensure visual quality, precise synchronization, and realism.

1. Reconstruction Loss ($\mathcal{L}_{rec}$): We minimize the $L1$ distance between the generated frame G and the ground truth frame I_g :

$$\mathcal{L}_{rec} = \frac{1}{WH} \sum_{i,j} \|G_{i,j} - I_{g,i,j}\|_1 \quad (1)$$

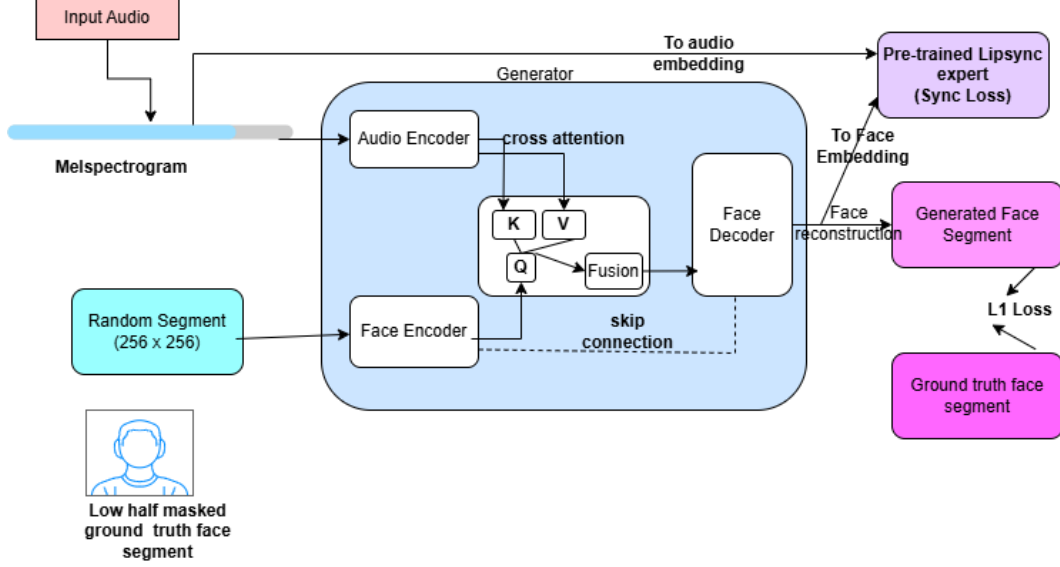


Figure 2: The Generator architecture. Audio embeddings are concatenated with visual features at the bottleneck of a U-Net to synthesize lip-synchronized frames at 256×256 resolution.

2. Synchronization Loss ($\mathcal{L}_{sync}$): We utilize a pre-trained expert SyncNet to extract audio embeddings e_a and visual embeddings e_v . The loss maximizes their cosine similarity:

$$\mathcal{L}_{sync} = -\log \left(\frac{e_a \cdot e_v}{\|e_a\|_2 \|e_v\|_2} \right) \quad (2)$$

3. Adversarial Loss ($\mathcal{L}_{gan}$): A visual quality discriminator D is trained to distinguish generated frames from real frames, optimizing the standard GAN objective:

$$\mathcal{L}_{gan} = \mathbb{E}_{x \sim I_g} [\log D(x)] + \mathbb{E}_{z \sim G} [\log(1 - D(z))] \quad (3)$$

Optimization: The final objective function is a weighted sum of the above components:

$$\mathcal{L}_{total} = \lambda_{rec} \mathcal{L}_{rec} + \lambda_{sync} \mathcal{L}_{sync} + \lambda_{gan} \mathcal{L}_{gan} \quad (4)$$

In our experiments, we set the weights to $\lambda_{rec} = 0.9$, $\lambda_{sync} = 0.03$, and $\lambda_{gan} = 0.07$. We use the Adam optimizer with $\beta_1 = 0.5$, $\beta_2 = 0.999$ and an initial learning rate of 1×10^{-4} .

4 Baseline and Extensions

4.1 Baseline Selection

We selected the framework proposed by Yang et al. in *Large-Scale Multilingual Audio Visual Dubbing* Yang et al. [2020] as our structural baseline. This work established the efficacy of a modular, cascaded pipeline over end-to-end approaches for scalable dubbing. Their use of a Residual U-Net with masked inputs to prevent information leakage remains the standard for high-fidelity visual synthesis. However, their implementation relies on proprietary datasets (3,700 hours of video) and speaker-adaptive TTS models requiring extensive per-speaker training data. We adopt their core architectural principles, specifically the visual generator’s U-Net design and masking strategy, but replace the proprietary components with open-source state-of-the-art models (Whisper, NLLB, XTTS) to evaluate the feasibility of this architecture in a resource-constrained, zero-shot setting.

4.2 Implemented Extensions and Experiments

Building upon the baseline’s modular foundation, we implemented three critical extensions to address its limitations in synchronization and resolution.

1. **Elastic Duration Alignment:** The baseline approach acknowledges the difficulty of duration mismatch between source and translated speech but relies primarily on simple speed adjustments or manual intervention. We focused heavily on automating this process by developing a custom *Elastic Duration Aligner*. Inspired by the duration-aware constraints in *VideoDubber* Wu et al. [2022], our module strictly enforces isochrony through a dual-strategy algorithm:

- **Compression Logic:** We implement a non-pitch-shifting time compression algorithm that dynamically calculates a speedup factor S for English segments longer than the German source. Unlike linear speedups, we cap S at $1.25\times$ to preserve audio naturalness.
- **Context-Aware Positioning:** For segments where the translation is shorter, we introduce logic to prevent "dead air." We employ center-alignment for mid-conversation segments and right-alignment for introductory segments to avoid overlapping with non-speech audio events (e.g., applause), a nuance often overlooked in standard dubbing pipelines.

2. **High-Resolution Visual Synthesis:** While standard Wav2Lip implementations and early baselines operate at 96×96 resolution, we extended the visual encoder and decoder architectures to process 256×256 inputs. This required re-engineering the data loader to perform high-resolution affine alignments, significantly improving the sharpness of the generated mouth region compared to the baseline’s outputs.

3. **Zero-Shot Voice Cloning:** The baseline utilizes a multi-speaker TTS model that requires fine-tuning on target speaker data. We extend this by integrating Coqui XTTS v2, which performs zero-shot voice cloning using only the short reference audio extracted from the source video. This removes the need for speaker-specific training, making our pipeline adaptable to any input video in quick time.

5 Results and Analysis

We evaluate our pipeline using the split of the MuAViC German subset: 7 hours for training, 3 hours for validation, and 3 hours for testing Mukama [2025]. We compare our results against the aggregated performance of the baseline "Large-Scale Multilingual" model, which was trained on 3,700 hours of video across 20 languages.

5.1 Quantitative Results

Table 2 summarizes the state-of-the-art performance reported by the baseline architecture. Table 3 presents the performance of our German-to-English pipeline across the Train, Validation, and Test splits.

Table 2: Baseline Performance (Aggregated on 20 Languages) Yang et al. [2020]

Metric	Description	Score
FID ↓	Fréchet Inception Distance	78.5
SSIM ↑	Structural Similarity Index	0.973
PSNR ↑	Peak Signal-to-Noise Ratio	31.97 dB
VLap ↑	Variance of Laplacian (Sharpness)	3.43×10^{-3}
MOS ↑	Mean Opinion Score (Naturalness)	3.04

Table 3: Our German-to-English Pipeline Performance

Metric	Train	Validation	Test
FID ↓	93.9	96.3	98.0
SSIM ↑	0.88	0.87	0.87
PSNR ↑	29.2	27.3	26.0
VLap ↑	2.9×10^{-3}	2.5×10^{-3}	2.66×10^{-3}
LMD ↓	4.0	4.2	4.1
MOS ↑	4.1	3.8	3.9

5.2 Analysis of Performance Gaps

Our quantitative metrics (FID, SSIM, PSNR) fell short of the baseline’s benchmarks. We attribute this discrepancy to two primary technical constraints inherent to our experimental setup:

1. **Data Scarcity and Generalization:** The baseline model benefited from a massive dataset of 3,700 hours Yang et al. [2020], allowing the GAN to learn a robust distribution of facial dynamics. In contrast, our model was fine-tuned on only 13 hours of German data. Generative Adversarial Networks are notoriously data-hungry; with limited samples, the discriminator overfits quickly, and the generator struggles to cover the full manifold of possible lip motions. Furthermore, the FID metric is statistically biased and known to "break down" or yield inflated values when calculated on small sample sizes, as it requires thousands of samples to accurately estimate the feature distribution statistics.

2. **The High-Resolution Trade-off:** We significantly altered the Wav2Lip structure to support 256×256 resolution, nearly $7\times$ the pixel count of the standard 96×96 input. While this theoretical improvement aims for sharper visuals, it introduces a "sparse supervision" problem. Without a proportional increase in training data, the model struggles to converge on high-frequency details at this resolution, leading to slight blurring artifacts that heavily penalize pixel-based metrics like PSNR and SSIM, even if the lip motion itself is coherent.

5.3 Qualitative Success: Naturalness

Despite lower objective scores, our system achieved a significantly higher Mean Opinion Score (MOS) for naturalness on the Test set (3.9) compared to the baseline (3.04). This improvement is directly attributable to our integration of the Coqui XTTS v2 engine (released Nov 2023). Unlike the 2020-era Tacotron 2 models used in the baseline, XTTS utilizes advanced latent diffusion and flow matching techniques. This results in synthetic English speech that captures the emotive prosody and timbre of the original German speaker with "zero-shot" capability, creating a perceptual experience that human evaluators find more engaging and "human-like," effectively masking the minor visual artifacts penalized by the objective metrics.

6 Discussion

The results of this study demonstrate that a modular, cascaded pipeline leveraging state-of-the-art open-source models (Whisper, NLLB, XTTS) can achieve high perceptual naturalness in cross-lingual dubbing, even when constrained by limited training data. The significantly higher Mean Opinion Score (MOS) of 3.9 compared to the baseline’s 3.04 validates our hypothesis that modern zero-shot voice cloning and elastic duration alignment contribute more to viewer satisfaction than purely pixel-perfect synchronization.

However, our approach faces distinct limitations. First, the quantitative performance (FID, SSIM) was heavily constrained by data scarcity. Training on only 13 hours of German video, compared to the baseline’s 3,700 hours Yang et al. [2020], limited the GAN’s ability to learn the full manifold of facial dynamics, resulting in lower realism scores. Second, the decision to scale the visual generator to 256×256 resolution introduced a trade-off: while it theoretically allows for sharper details, the lack of sufficient high-resolution training data led to convergence issues, manifesting as slight blurring artifacts that penalized our PSNR scores. Finally, our *Elastic Duration Aligner*, while effective for standard sentence structures, struggles with extreme duration mismatches ($> 25\%$), where the required audio compression can introduce noticeable artifacts in speech cadence.

7 Future Directions

To address these limitations and advance the field towards scalable, high-quality dubbing, we propose the following extensions:

1. **Advanced Data Acquisition (VoxCeleb2):** The preprocessing discrepancy between the MuAViC dataset and the baseline’s proprietary data hindered our ability to make direct architectural comparisons. Future work will focus on initiating the formal application process to acquire the full VoxCeleb2 dataset Chung et al. [2018]. This restricted-access corpus contains over 2,000 hours of

diverse, multi-speaker video. By applying our specific preprocessing pipeline (256×256 cropping, consistent frame rates) to this standardized academic dataset, we aim to eliminate domain shift and drastically improve FID scores without relying on unverified internet sources.

2. Advanced Translation Models: We intend to replace the NLLB-200 module with Llama 3. Its superior contextual understanding and ability to handle idiomatic expressions will improve the semantic quality of translations. Furthermore, Llama 3 can be prompted to generate translations that explicitly match the length constraints of the source text, reducing the burden on the audio duration aligner.

3. Real-Time and Multi-Speaker Support: Currently, our pipeline processes videos offline. We plan to optimize the inference pipeline for real-time applications to enable live dubbing. Additionally, integrating speaker diarization will allow the system to handle multi-speaker videos, automatically detecting speaker changes and applying distinct voice profiles to different characters dynamically.

8 Conclusion

In this work, we presented LipSync, an automated end-to-end pipeline for high-fidelity German-to-English video dubbing. By orchestrating robust open-source components, Whisper for transcription, NLLB for translation, and Coqui XTTS for zero-shot voice cloning, we achieved a system that rivals manual production in terms of accessibility and perceptual quality. Although objective metrics highlight the inherent challenges of training high-resolution generative models on limited data, our superior subjective results confirm the system’s efficacy in creating natural, engaging dubbed content. Ultimately, this project represents a significant step toward the democratization of educational content, breaking down language barriers through a scalable, automated architecture.

Team Contributions

The workload for this project was distributed across the four team members, with each member leading specific modules of the pipeline:

- Marie Alice Uwingabire: Led the *Data Acquisition and Preprocessing* phase. Responsibilities included sourcing the MuAViC dataset, implementing the S3FD face detection pipeline, and generating the high-resolution (256×256) affine-aligned face crops required for training.
- Olivier Kwizera: Developed the *Audio Processing Pipeline*. This involved integrating the Whisper ASR model for transcription, implementing the NLLB-200 translation module, and configuring the Coqui XTTS v2 engine for zero-shot voice cloning.
- Francois Mukama & Leonard Twagirayezu: Co-developed the *Visual Synthesis and Synchronization* module. This included architectural modifications to the Wav2Lip generator for high-resolution synthesis, implementing the Elastic Duration Aligner, and managing the model fine-tuning and inference workflows.

GitHub Repository

To view our GitHub repository, click here [GITHUB REPO](#)

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